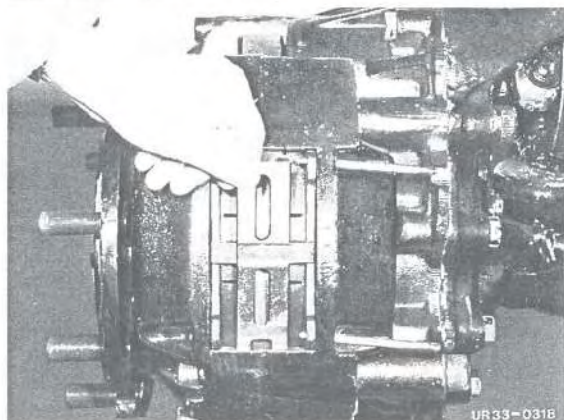
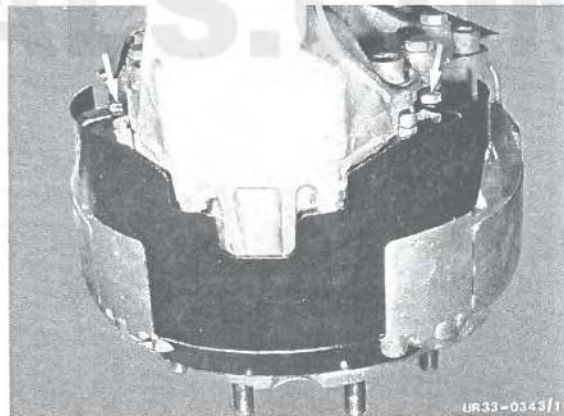
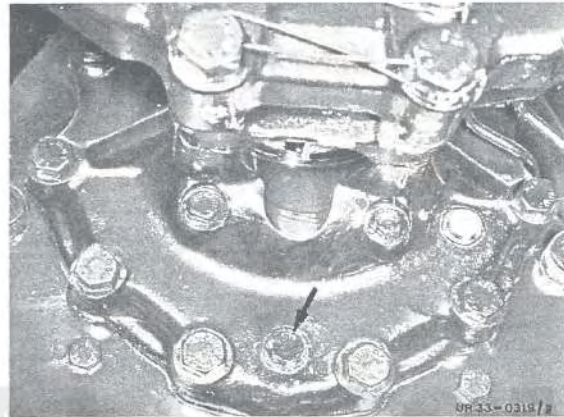
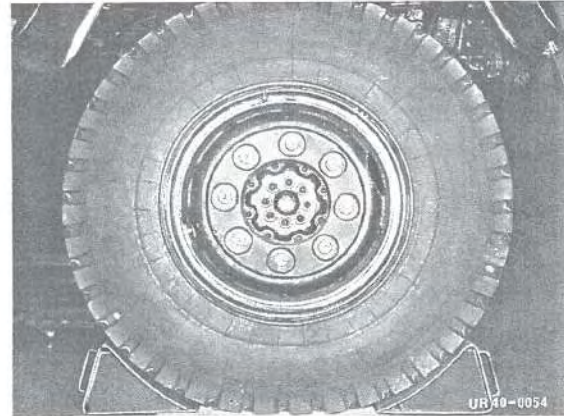


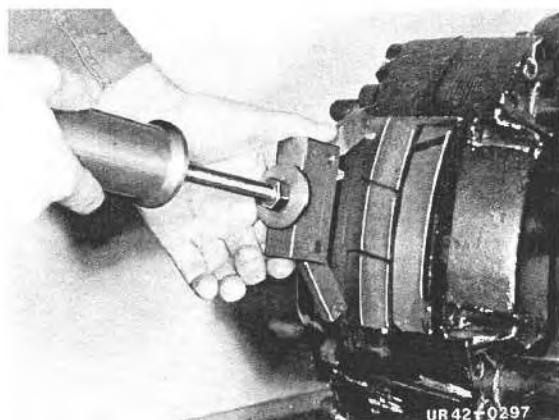
Disassembly

Note: Operations at left and right-hand wheel hub drives are to be executed in same order. Work is same for axle in situ.

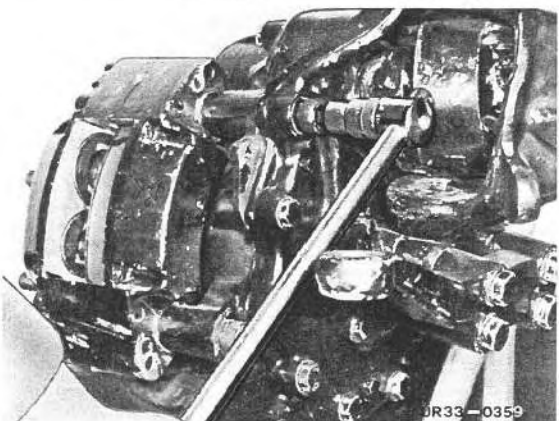
- 1 Remove front axle, referring to 2.1/1.
- 2 Detach wheels.
- 3 Drain oil off wheel hub drive.
- 4 Remove brake backplate.
- 5 Unscrew fixed brake caliper line and bleeder line, sealing all ends.
- 6 Detach backplate from fixed caliper, knock pins out and remove spring clips.



737.2

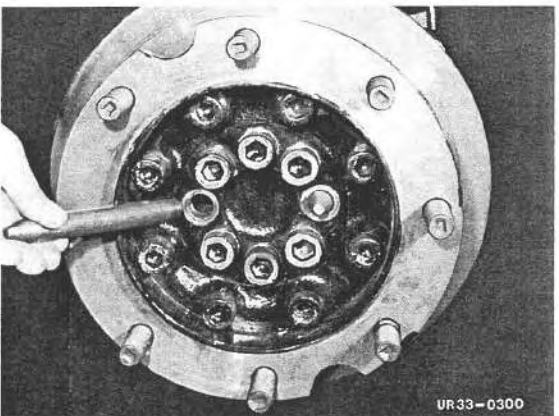


7 Remove brake pads using special tool No. 12.



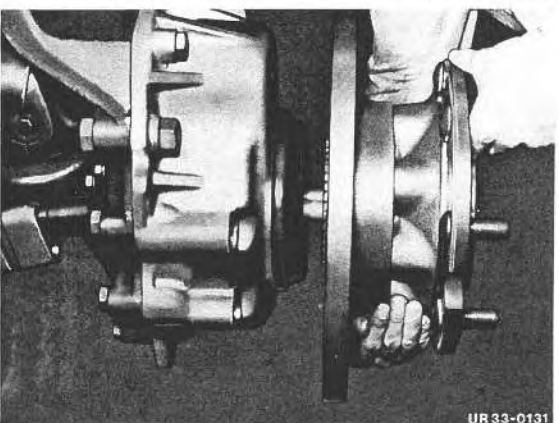
8 Unscrew fixed caliper using special tool No. 4.

9 Unscrew two wheel hub fastening bolts, using special tool No. 3.

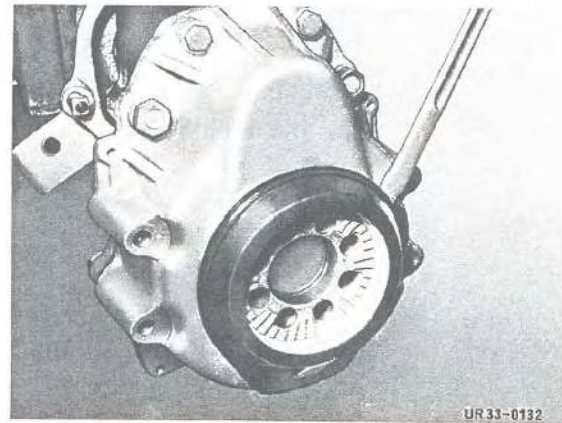


10 Screw special tool No. 6 into position and unscrew all other fastening bolts.

11 Remove wheel hub with brake disk.



12 Remove oil baffle.

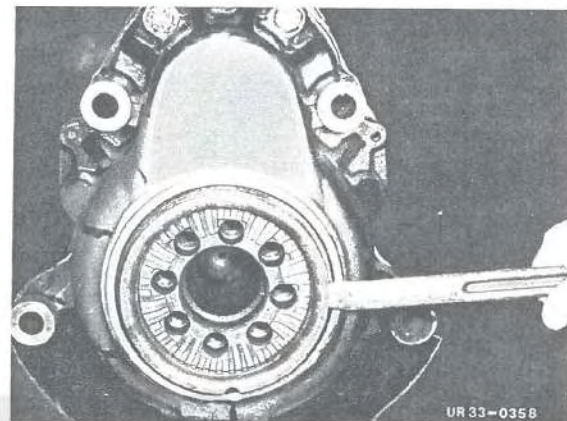


UR33-0132

13 Remove wear ring.

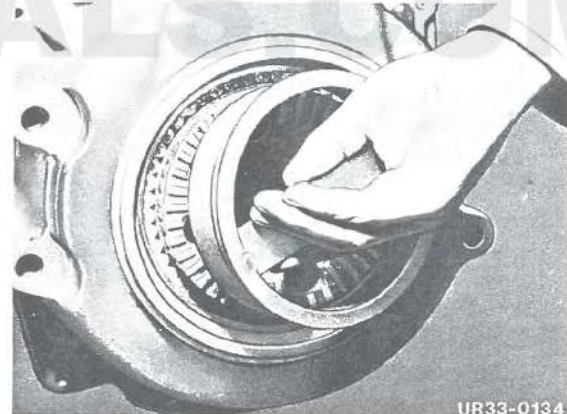
Note: In order to avoid damage, do not tilt when removing the wear ring.

14 Remove shaft sealing ring.



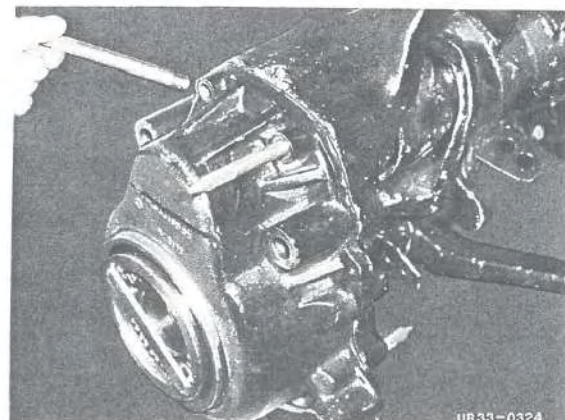
UR 33-0358

15 Insert special tool No. 4 in roller bearing.



UR33-0134

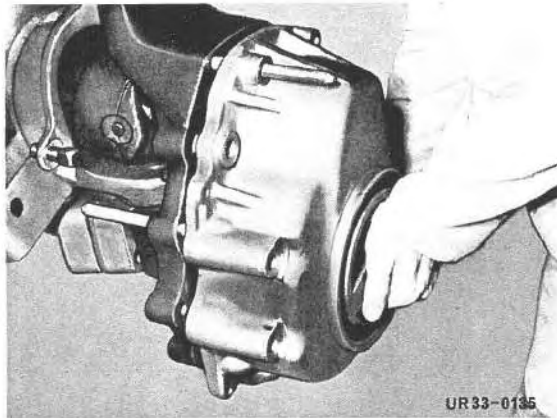
16 Release wheel hub drive housing and screw special tool No. 6 in position.



UR33-0324

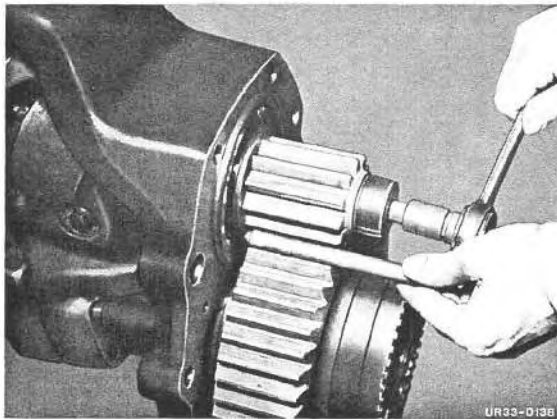
33.3 Disassembly and Assembly of Wheel Hub Drive

737.2



17 Remove housing, and at the same time slip special tool No. 5 completely into the roller bearing so that the rollers cannot fall out.

Note: Special tool No. 4 is held by the O-ring so that it cannot fall out of the bearing.

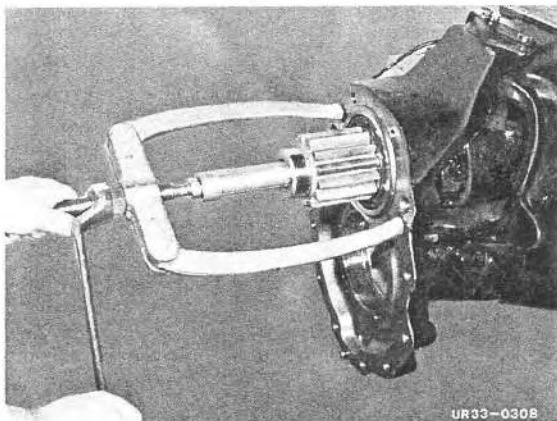


18 Block drive gear with respect to driven gear (e.g. by inserting a copper drift). Unscrew center fastening bolt using special tool No. 3.

19 Release fastening bolts of the bearing ring at the steering knuckle. Remove driven gear using special tools No. 11 and No. 13.

Caution - Danger

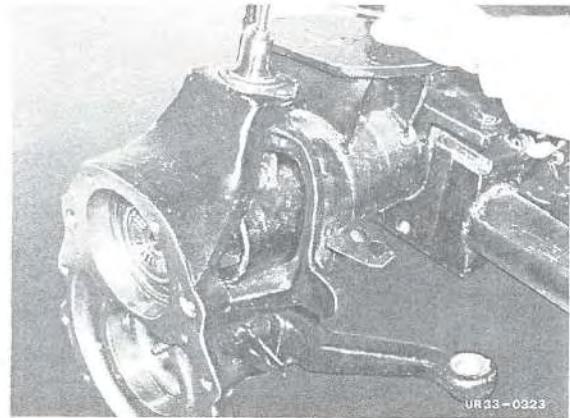
Attach special tools to crane since driven gear may otherwise fall down.



20 Detach drive gear using puller and special tool No. 11.

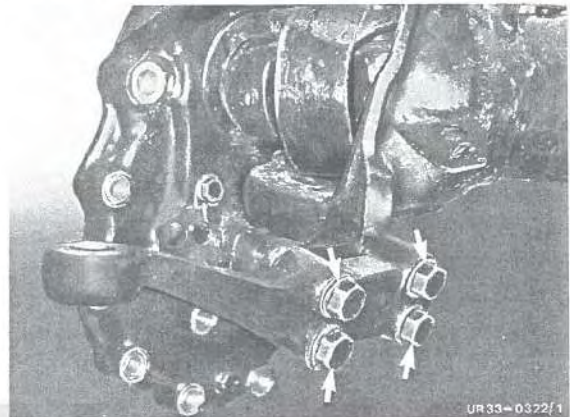
737.2

21 Remove end plate. Pull upper swivel joint off, using special tool No. 9.

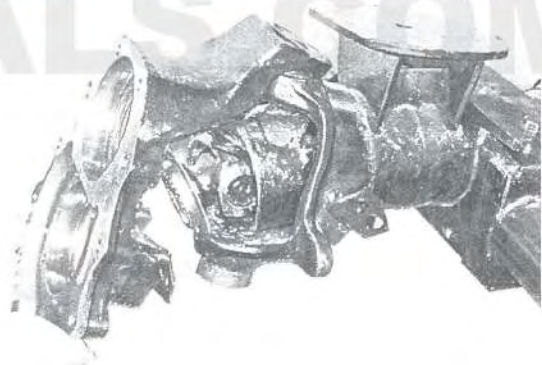


22 Unscrew fastening bolts and remove steering knuckle arm and/or steering arm together with shell bearing half, thrust piece and thrust shell.

Note: These arms must only be unscrewed after axle has been jacked up because fastening bolts also hold lower swivel joint in addition to bearing cap.



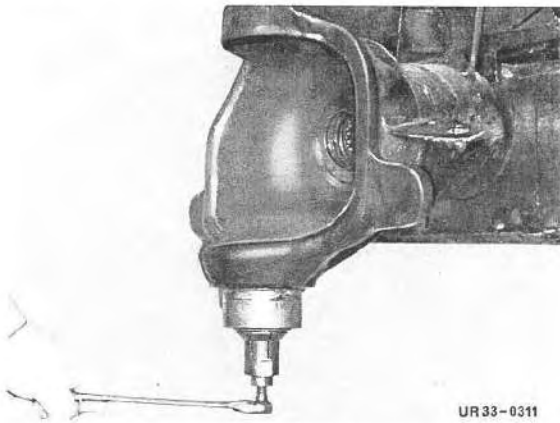
23 Remove steering knuckle.



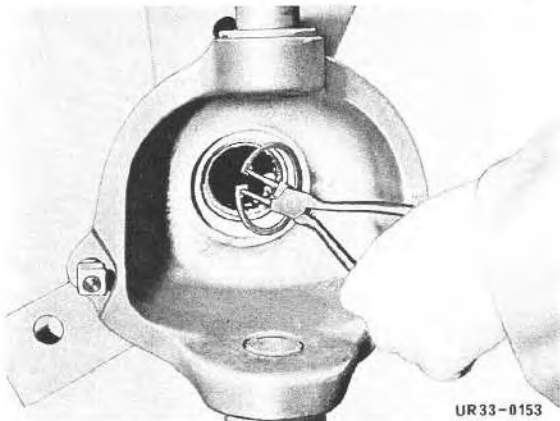
24 Remove constant-velocity joint shaft.



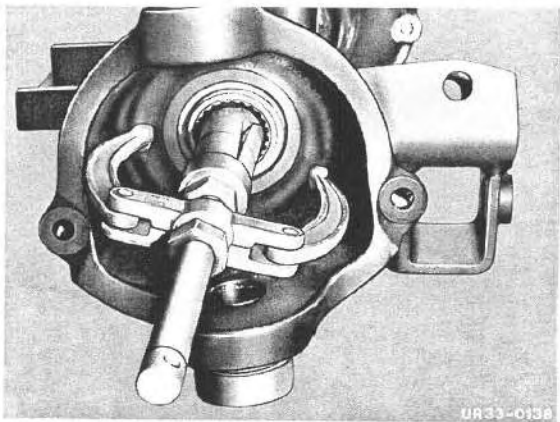
737.2



25 Withdraw lower joint bearing, using special tool No. 9.



26 Check condition of needle bearing in axle tube, remove shaft seal and snap ring if necessary.

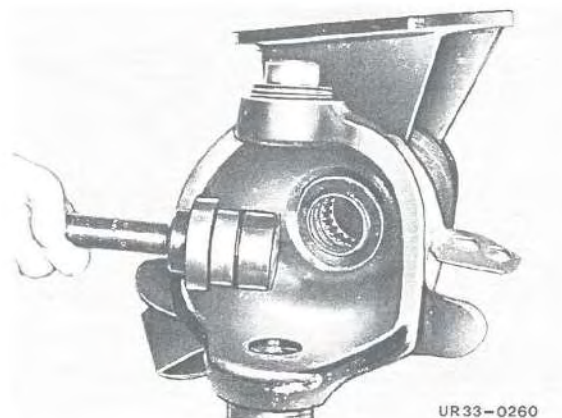


27 Withdraw needle bearing, using puller and special tool No. 11. Be sure not to damage snap ring beyond bearing.

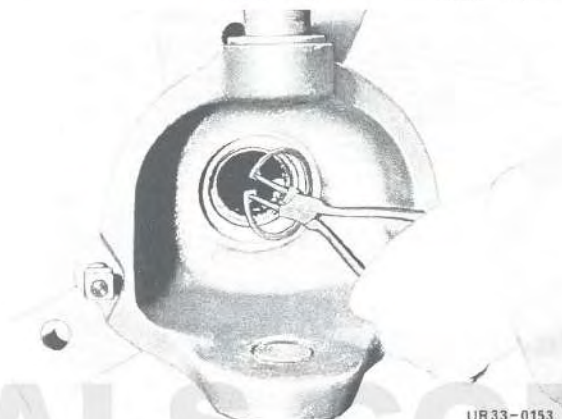
28 Clean and check all parts, exchange if necessary.

Assembly

1 Using special tool No. 16, force needle bearing into axle suspension tube as far as rear snap ring. Coat with fluid grease No. 2.



2 Fit front snap ring.

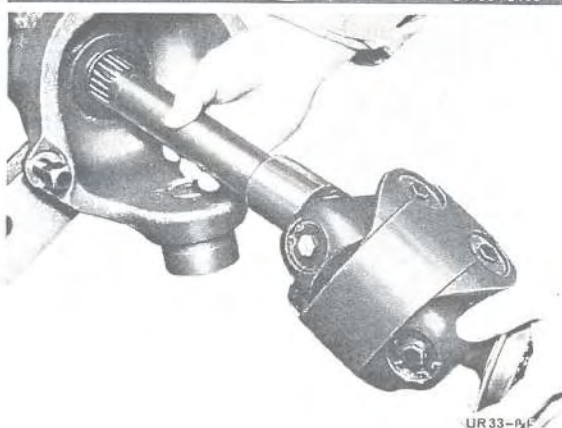


3 Coat outside of shaft seal with sealant No. 5 and press into position using special tool No. 16.

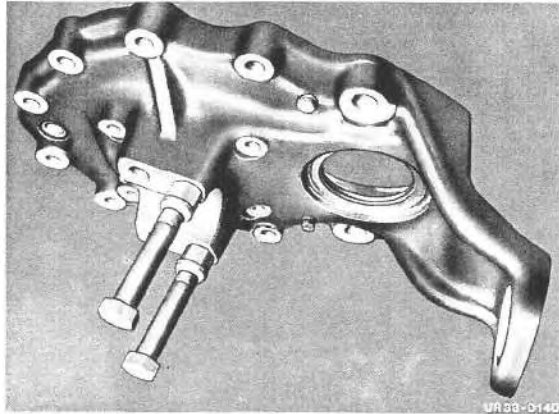


4 Coat splining of constant-velocity joint shaft with paste No. 3 prior to insertion. Coat inside diameter of wear ring with sealant No. 5.

Note: If necessary, exchange wear ring and protective plate of shaft seal on constant-velocity joint shaft.



737.2



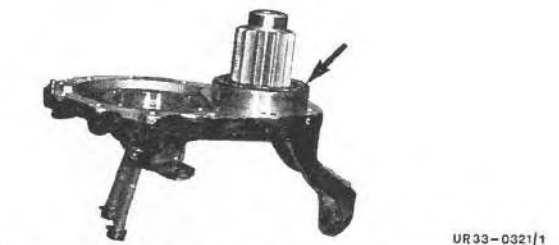
5 Screw two bolts and nuts into steering knuckle.



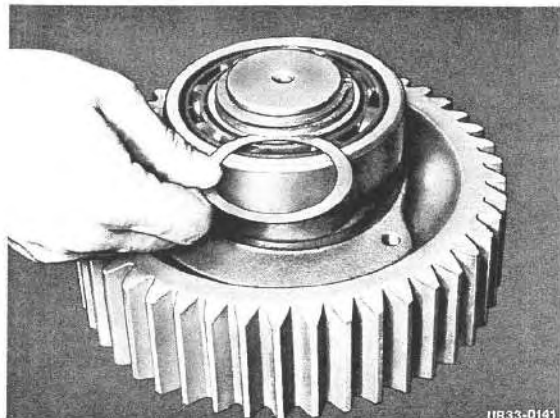
6 Position steering knuckle at horizontal angle.

7 Coat shaft seal with lubricating grease No. 2 and force into position using special tool No. 15.

8 Press deep-groove ball bearing onto drive gear and force into steering knuckle.

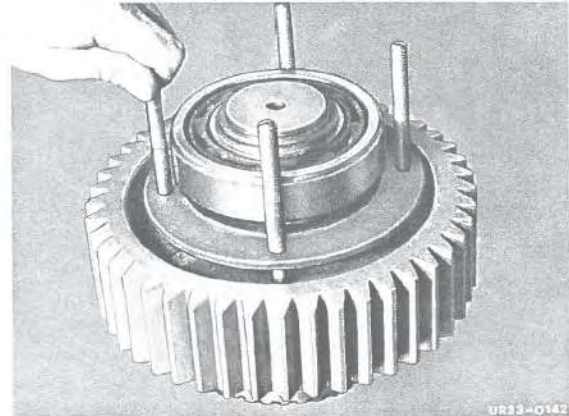


9 If deep-groove ball bearing is worn at driven gear side, insert bearing ring and fit deep-groove ball bearing.



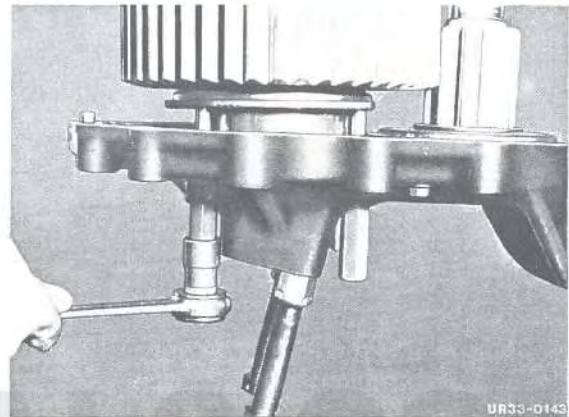
10 Insert supporting washer and snap ring, obtaining distance "O" between supporting washer and driven gear by using suitable shim.

- 11 Screw special tool No. 10 into bearing ring.

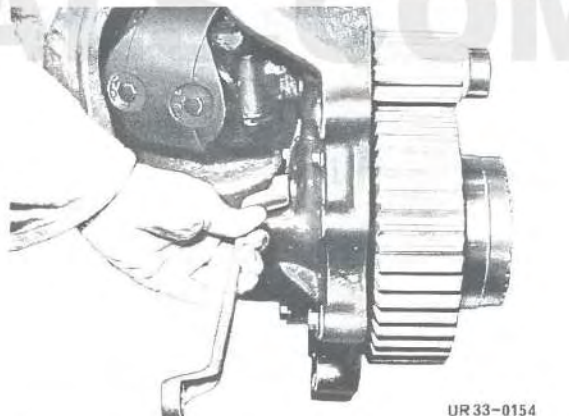


- 12 Using special tool No. 10, draw driven gear evenly into bearing seat.

Note: Make sure that contact flange of bearing ring is pointing toward deep-groove ball bearing of drive gear.



- 13 Coat fastening bolts with sealant No. 1 and tighten driven gear.

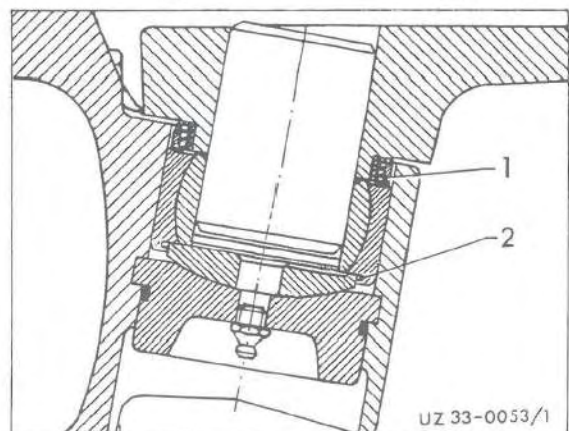


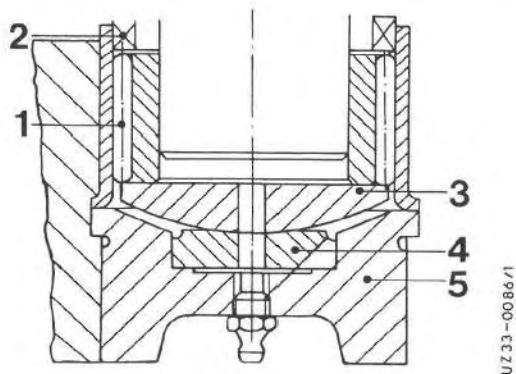
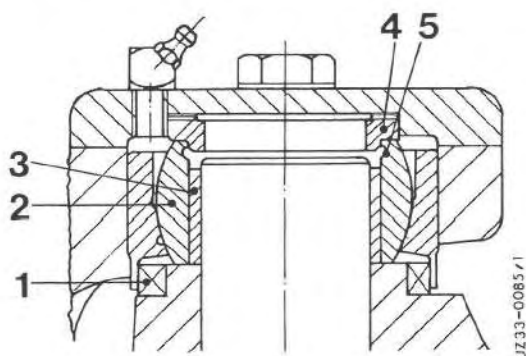
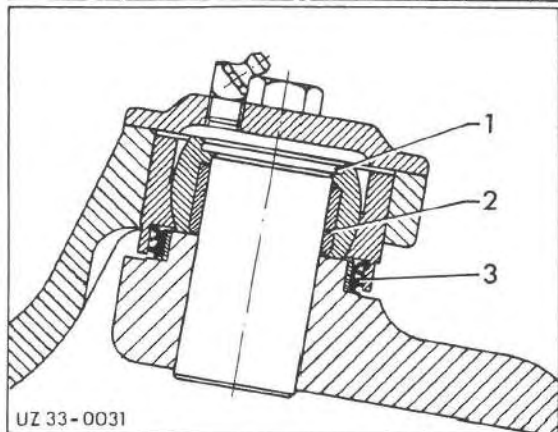
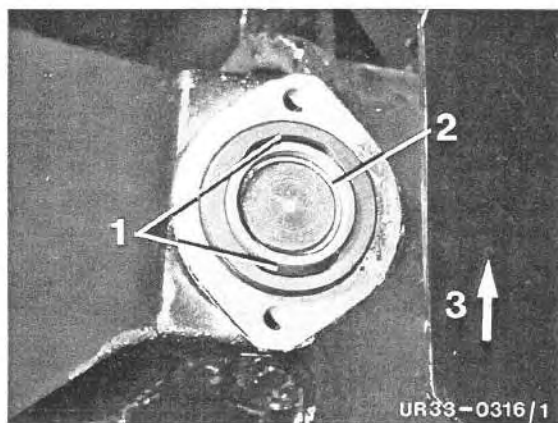
- 14 Coat shaft seal (1) with lubricating grease No. 2 and force into lower swivel joint. Press swivel joint onto kingpin.

Note: Extracting groove (2) of swivel joint must point downwards for bearing removal. Recess in outer race must point in straight-ahead direction.

From Model	Chassis end No.	Front axle No.
435.111	002 783	002 372

no more extension tube on lubricating nipple and no more bore in kingpin.





15 Hang preassembled steering knuckle over steering knuckle pin.

16 Coat shaft sealing ring with grease No. 2 and press into upper pivotted bearing. Press pivotted bearing onto steering knuckle pin.

Recess on outer race must point in direction of travel (3).

Note: To enable pivotted bearing (2) to be removed, install removal slot (1) of pivotted bearing upward.

17 Screw closing plate onto steering knuckle.

18 Coat bearing half shell with sealing compound No. 1. Tighten thrust piece, thrust shell and bearing half shell with steering knuckle arm or track rod arm and secure.

Note: Model 435.110/111 from Chassis end No.
113 086 509
086 019

(steering knuckle with needle bearing) assemble steering knuckle bearing top or bottom as follows:

19 Coat shaft sealing ring (1) with grease and press into upper pivotted bearing.

Coat pivotted bearing (2) with running-in paste No. 3 and press together with bush (3) onto steering knuckle pin.

Note: To enable the pivotted bearing (2) to be removed, install extractor rivet (5) of pivotted bearing upward.

20 Hand preassembled steering knuckle over steering knuckle pin.

21 Coat shaft sealing ring (2) with grease No. 2 and install.

23 Coat bearing half shell and fastening bolts with sealing compound No. 1. Tighten needle bearing (1), thrust piece (3), fulcrum pad (4), thrust shell (5) and bearing half shell with steering knuckle arm and secure. Refer to 1.5/1 for tightening torques.

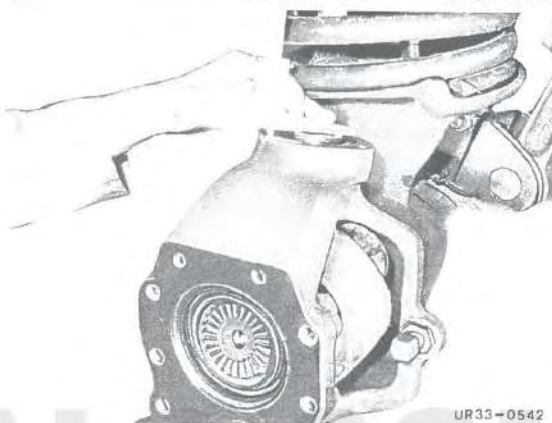
24 Calculate installation depth for spacer ring (4) in closing plate.



25 Install spacer ring (4) into pivoted bearing. Compensate gap between the two calculated dimensions plus permissible pretension with shims. Refer to 1.3/1 for permissible pretension. Compensating may only be performed when the axle is not subjected to load.

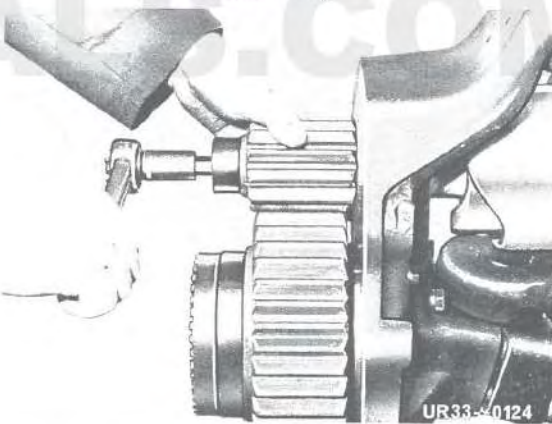
Note: If compensating is performed with the wheel hub drive fitted, support hub drive from below.

26 Bolt closing plate onto steering knuckle.

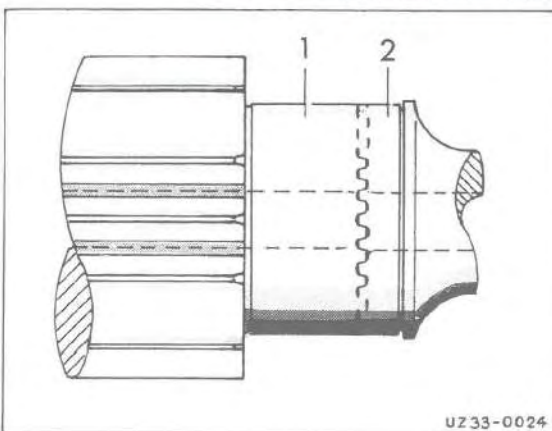


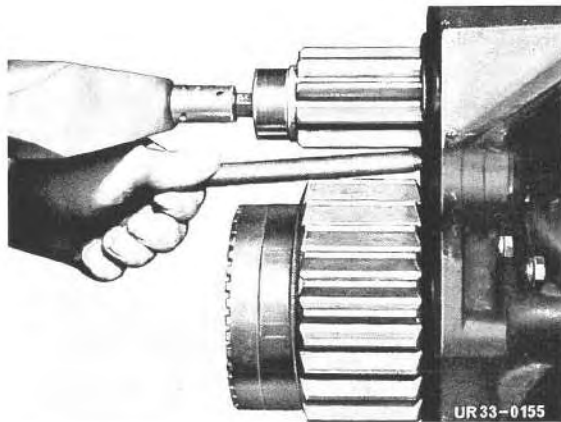
27 Bolt on drive gear with special tool No. 3.

Note: Screw in central fastening bolt, turning drive gear at the same time until a backlash is detectable.



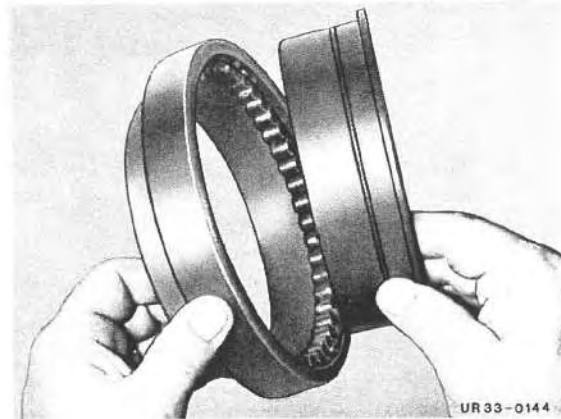
28 Turn fastening bolt through further 2 to 2 1/2 turns and bolt drive gear (1) with double propeller shaft (2) tooth on tooth gap.



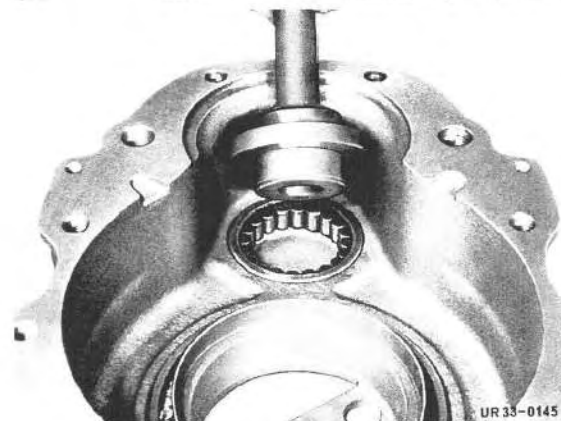


29 Lock drive gear to output gear (e.g. by inserting a copper drift). Tighten central fastening bolt with special tool No. 3.

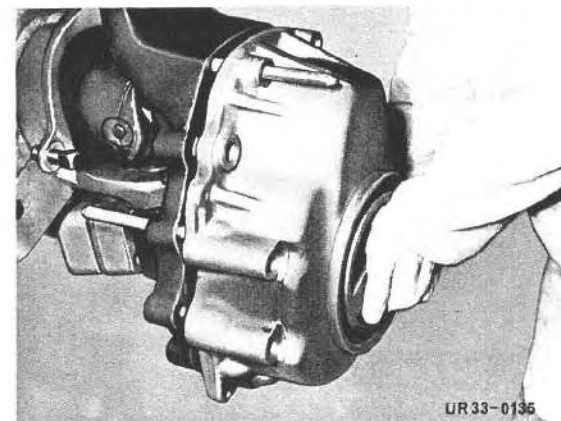
Refer to 1.5/1 for tightening torques.



30 When installing the large roller bearing on the output gear, replace transportation sleeve with special tool No. 5. Press roller bearing in with special tool No. 14 and coat with fluid grease No. 2.



31 If a small roller bearing in the hub drive housing was defective, press new roller bearing into hub drive housing with special tool No. 23.



32 Screw in special tool No. 6.

33 Coat contact surface of hub drive housing with sealing compound No. 1 and fit housing.

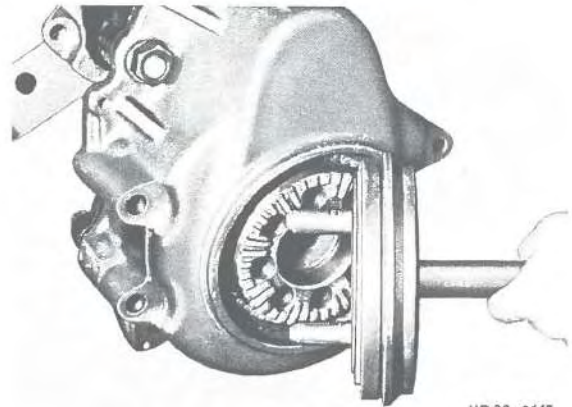
Note: Pay attention to special tool No. 4 to avoid any roller bodies falling out.

34 Coat shaft sealing ring with grease No. 2 and install with special tool No. 14.

Note:

From Model	Chassis end No.	Front axle No.
435.111	071 365	

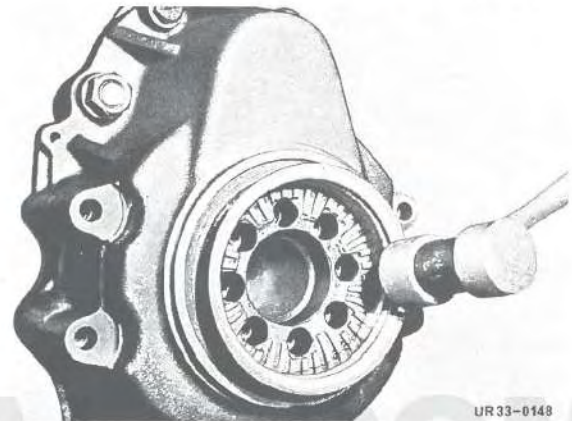
New shaft sealing ring with modified lip and new material for better seal installed.



UR 33-0147

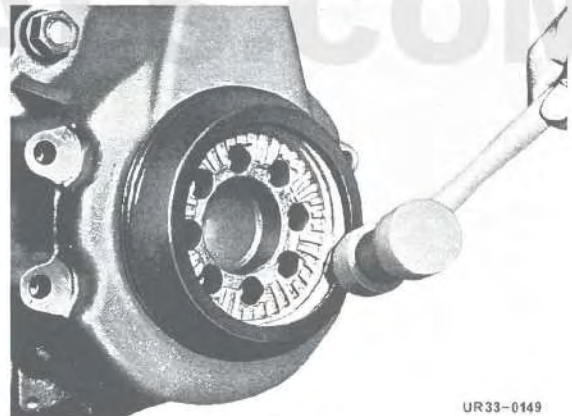
35 Install wear ring.

Note: If dust lip of shaft sealing is pressed into wear ring, pull out dust lip with wire.



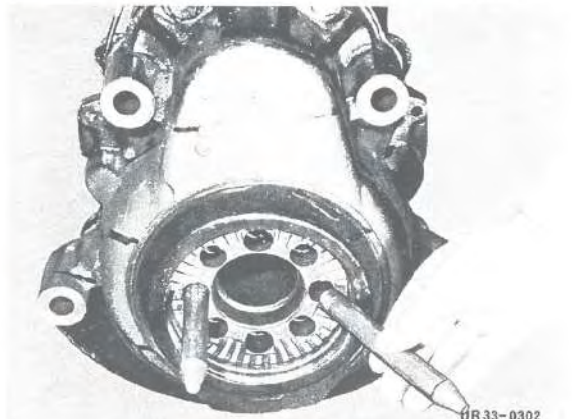
UR 33-0148

36 Fit oil baffle.

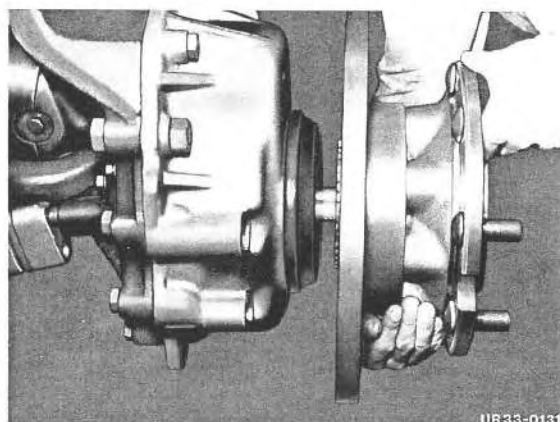


UR 33-0149

37 Screw special tool No. 5 into drive gear.



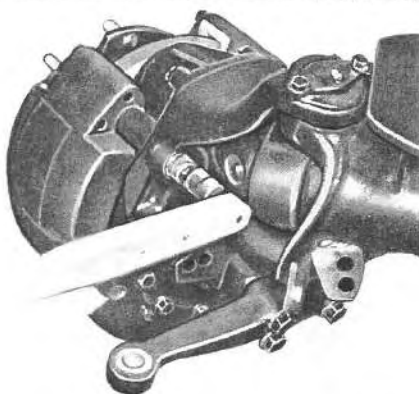
UR 33-0302



38 Position sealing ring on wheel hub.

39 Slip wheel hub and brake disk over special tool No. 6 and torque.

See 1.5/1 for tightening torques.



40 Attach fixed caliper and torque, using special tool No. 4.

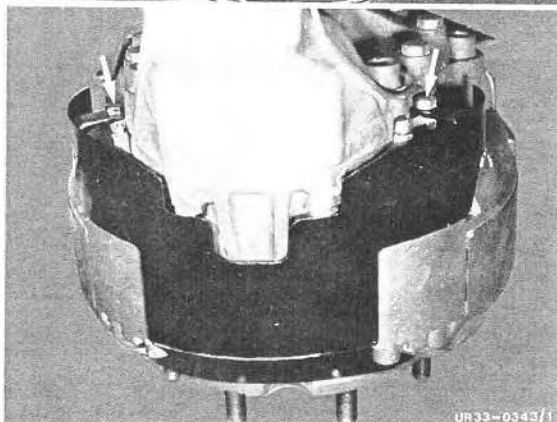
See 1.5/1 for tightening torques.

41 Connect brake lines to caliper and bleeder line to steering knuckle.



42 Install brake pads.

43 Fill wheel hub drive with oil, referring to 1.1/2 for quantity.



44 Attach brake backplate.

45 Install front axle.

46 Attach wheels and torque wheel nuts.
See 1.5/1 for tightening torques.

Note: Retorque wheel nuts after 50 km.

47 Bleed brake system and check for leakage.

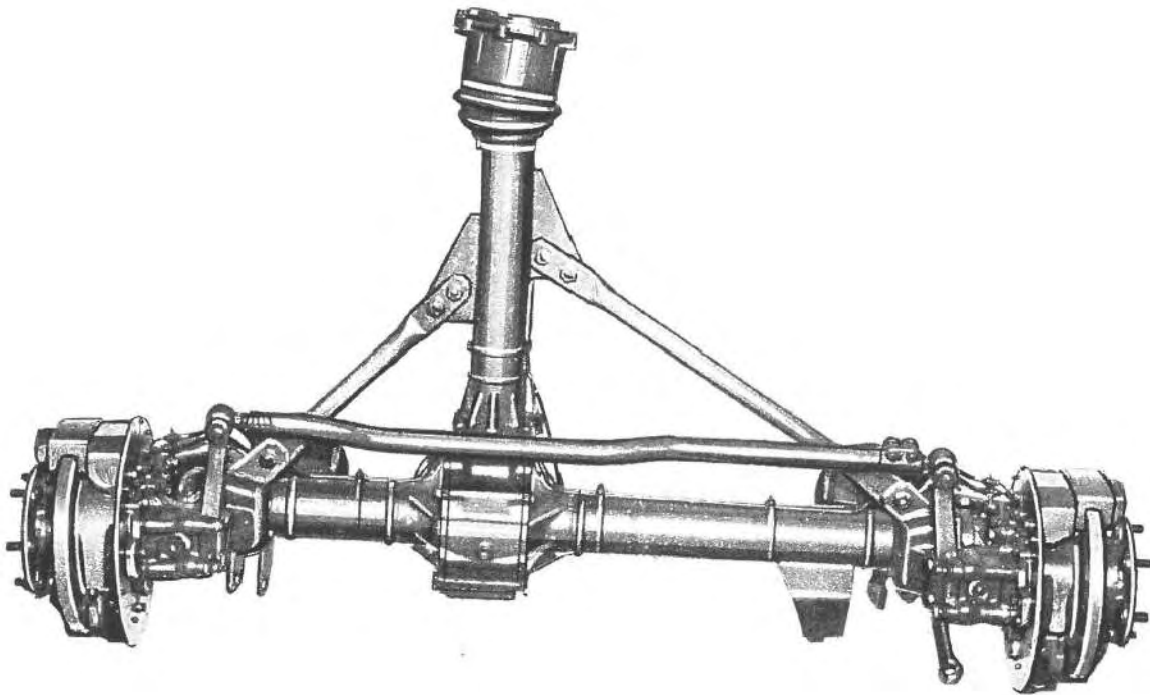
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Worksheet for Adjusting Axle Drive	–
2 Removal and Installation of Front Axle	2.1/1
3 Disassembly and Assembly of Front Axle	3.1/1
4 Disassembly and Assembly of Drive Pinion Bearing	4.1/1
5 Disassembly and Assembly of Differential Gearing	5.1/1
6 Disassembly and Assembly of Wheel Hub Drive	6.1/1
7 Checking Differential Lock for Leaks	7.1/1

Installation Survey

Chassis		Front axle		Brake	Installation
Model	Sales designation	Model	Sales designation		
435.115	U 1300 L	737.111	AU 2/14 S - 4,0	Disk brake	Series
435.117	U 1300 L/37				

Overall view



UR33-0550

Front axle

33.6 General

737.111

Technical Data

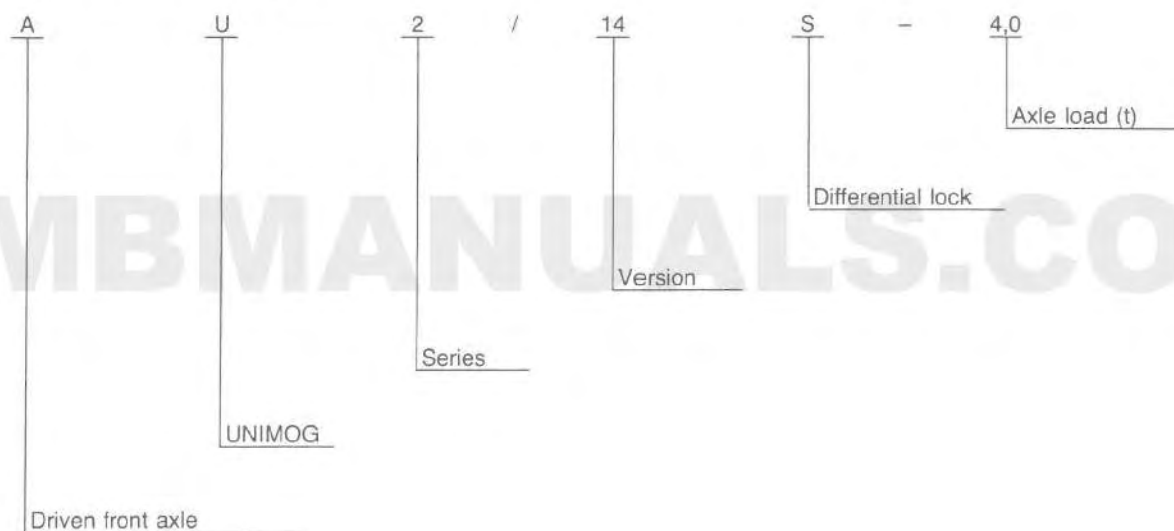
Ratios

Axle model	Installation	Gear train Ring gear drive pinion		Gear train Wheel hub drive		Total Ratio
		Number of teeth	Ratio	Number of teeth	Ratio	
737.111	Series	22: 7	3,143	27: 13	2,077	6,527
737.111	SA	23: 9	2,555			5,307

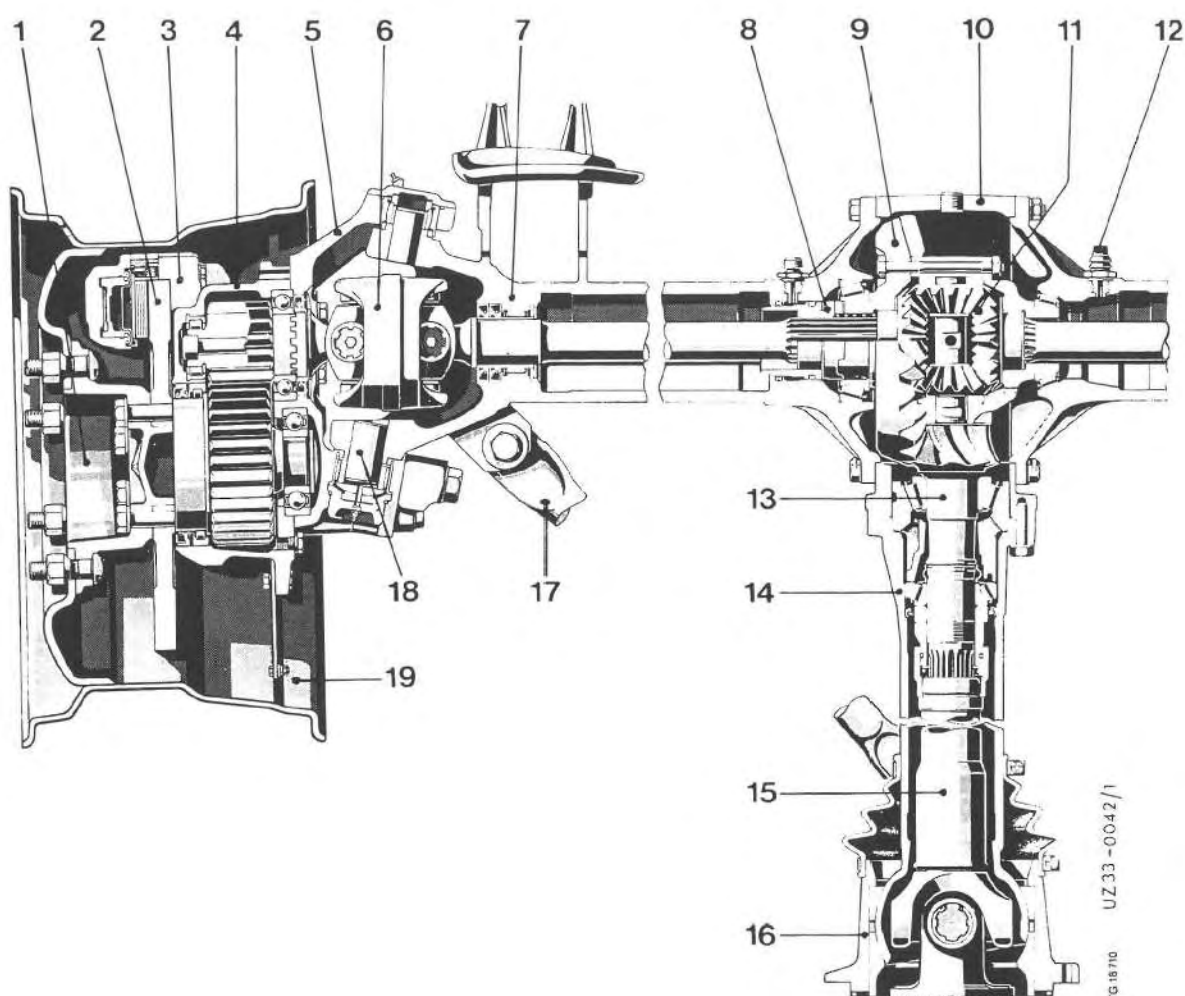
Capacities

Designation	Service product	SAE class	Capacity (L)	Axle model
Axle center housing	Gear oil	90	2,5	737.111
Wheel hub drive			each 0,3	

Key to Front Axle Designation



Sectional View



AU 2/14 S – 4.0

- | | | | |
|----|--|----|-------------------------------|
| 1 | Wheel hub | 11 | Differential |
| 2 | Brake disk | 12 | Axle ventilation connection |
| 3 | Fixed caliper | 13 | Drive pinion |
| 4 | Wheel hub drive | 14 | Torque tube |
| 5 | Steering knuckle, outside | 15 | Drive shaft |
| 6 | Constant-velocity joint shaft | 16 | Torque ball casing with shell |
| 7 | Axle tube with steering knuckle inside | 17 | Axle strut |
| 8 | Differential lock | 18 | Kingpin |
| 9 | Ring gear | 19 | Disk wheel |
| 10 | Center axle housing | | |

UZ 33-0042/1

TG 18 710

Setting Values

Designation	Setting Value
Preload of torque ball shells on extension tube mounting flange manual effort measured.	200 to 300 N
Breakaway force for drive pinion, refer to axle data sheet	3,0 to 3,5 Nm
Backlash in differential	0,15 to 0,20 mm
Claw play of differential lock, refer to axle data sheet	0,5 ± 0,2 mm
Axle play of bearing on output gear	0
Axle play of bearing in wheel hub drive	0 No preload
Bearing play of tapered roller bearing on differential	0 No preload
Preload, steering knuckle bearing	0,2 to 0,3 mm
Adjustment A for basic setting of steering	165 mm

Necessary Materials

Cons. No.	Designation	Part Number
1	Teroson Atmosit sealing compound	000 989 43 20 ¹⁾
2	Longlife grease	commercially available
3	Molybdenum disulphide running-in paste	commercially available
4	Loctite No. 241 adhesive	002 989 70 71
5	Loctite No. 270 adhesive	002 989 93 71
6	Molykote liquid grease	commercially available
7	Anti-corrosive oil SAE 30	commercially available
8	Hypoid gear oil SAE 90	commercially available
9	Loctite No. 573	001 989 46 20
10	Teroson - Fluid	000 989 05 20
11	Terostat 56 surface sealing compound	001 989 58 20

1) optional 001 989 58 20

General note: In order to ensure satisfactory adhesion of sealing compounds, the sealing surfaces must be completely free of oil and grease.

Special Tools

Cons. No.	Designation	Special Tool No.	Tool set
1	Extension tube for mounting flange	000 589 01 14 00	A
2	14 mm pin wrench socket, 3/4 inch square for drive gear	001 589 68 09 00	B
3	24 mm socket for wheel hub 3/4 inch square for wheel hub	001 589 71 09 00	B
4	Assembly drift (tool required) for wheel hub and break disk	424 589 00 15 00	B
5	Drift for installer	406 589 04 15 00	B
6	Locking fixture for drive gear	435 589 01 31 00	B
7	Puller for plain bearing	424 589 00 33 00	B
8	Puller for drag link and track rod	406 589 05 33 00	B
9	Counter support for extractor	000 589 35 33 00	B
10	Extractor for break pads	001 589 43 33 00	B
11	Installer for shaft seal in torque tube	435 589 01 35 00	B
12	Installer for shaft seal in steering knuckle and axle tube	424 589 00 43 00	B
13	Installer for roller bearing and shaft seal in wheel hub drive	424 589 01 43 00	B
14	Piston resetting fixture for brake piston	000 589 52 43 00	B
15	Mounting flange for torque tube (with tool No. 1)	425 589 00 59 00	B
16	Fitting sleeve for torque tube	406 589 01 59 00	B
17	Adjusting key for drive pinion	000 589 64 03 00	C
18	Installer for shaft seal rings in axle tube	387 589 01 15 00	C
19	Installer for needle bearings in axle tube	387 589 02 15 00	C
20	Adjuster for drive pinion	425 589 00 21 00	C
21	Torque wrench for drive pinion (0 to 5 Nm) 1/4 inch square	000 589 87 21 00	C
22	Pressure tester for differential lock (checking for leaks)	401 589 01 25 00	C
23	Mount for drive pinion	425 589 00 31 00	C
24	Installer for taper roller bearings in axle tubes and differential (with tool No. 6)	406 589 02 43 00	C
25	Installer for taper roller bearings on differential (with tool No. 6)	406 589 03 43 00	C
26	Installer for small roller bearing in wheel hub drive	406 589 05 43 00	C

Tightening Torques

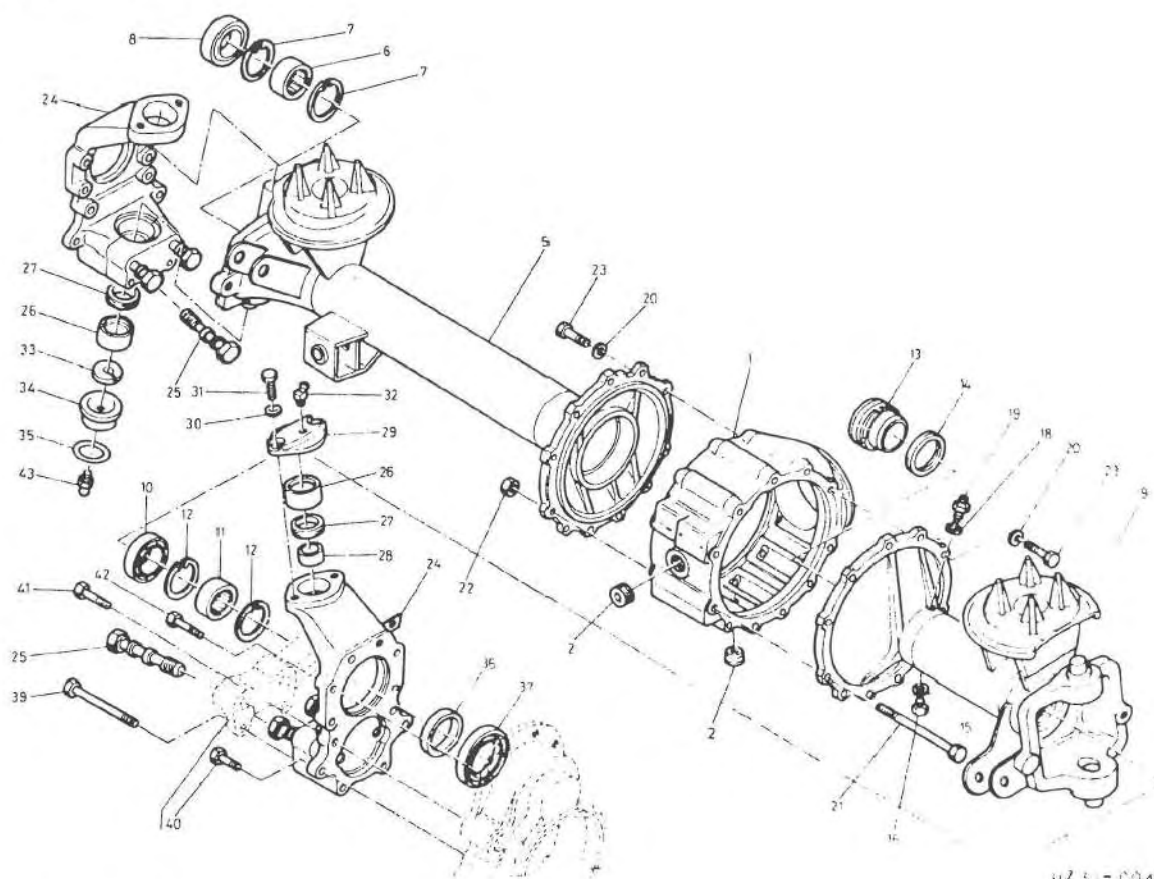
Designation	Thread	Nm
Wheel hub to drive gear	M 16 x 1,5– 12.9	320 ± 20
Nut clamping bolt output gear	M 20 x 1,5– 10.9	220
Drive gear to drive shaft	M 16 x 1,5– 12.9	300
Fixed caliper to wheel hub drive housing	M 16 x 1,5– 10.9	250
Drag link or track rod arm steering knuckle ¹⁾ (microencapsulated)	M 12 x 1,5– 10.9	110
Drag link or track rod arm to steering knuckle (necked-down bolts)	M 20 x 1,5– 12.9	450
Shock absorber mounting to axle	M 20 x 1,5– 8.8	250
Shock absorber mounting to frame	M 20 x 1,5– 8.8	250
Axle strut to axle tube	M 18 x 1,5– 10.9	350
Axle strut to torque tube	M 18 x 1,5– 10.9	350
Transverse link to frame	M 16 x 1,5– 10.9	315
Transverse link to axle	M 16 x 1,5– 10.9	315
Front spring to frame	M 18 x 1,5– 8.8	180
Drag link or track rod to steering arm and/or track rod arm	M 20 x 1,5	230 ± 20
Clamp to steering arm and track rod	M 12 x 1,5– 8.8	90 ± 3
Wheel stop screw to steering knuckle, inner	M 16– 8.8	200
Axle tube assembly to axle center housing	M 14 x 1,5– 12.9	200
Ring gear to differential case	M 12 x 1,5	120
Torque tube to axle center housing	M 12 x 1,5– 10.9	90
Wheel nuts ²⁾	A 18	400

1) From chassis end No. 108 223 or from front-axle No. 900 78 33 (steering knuckle arm)
 From chassis end No. 108 620 or from front-axle No. 900 82 43 (track rod arm)
 Reinforced in connection with 2 socket head bolts M 12 x 1.5 x 155 microencapsulated.

2) For type 435.117 with rim 11 - 20 SDC (hub centering) 450 Nm.

737.111

Exploded View



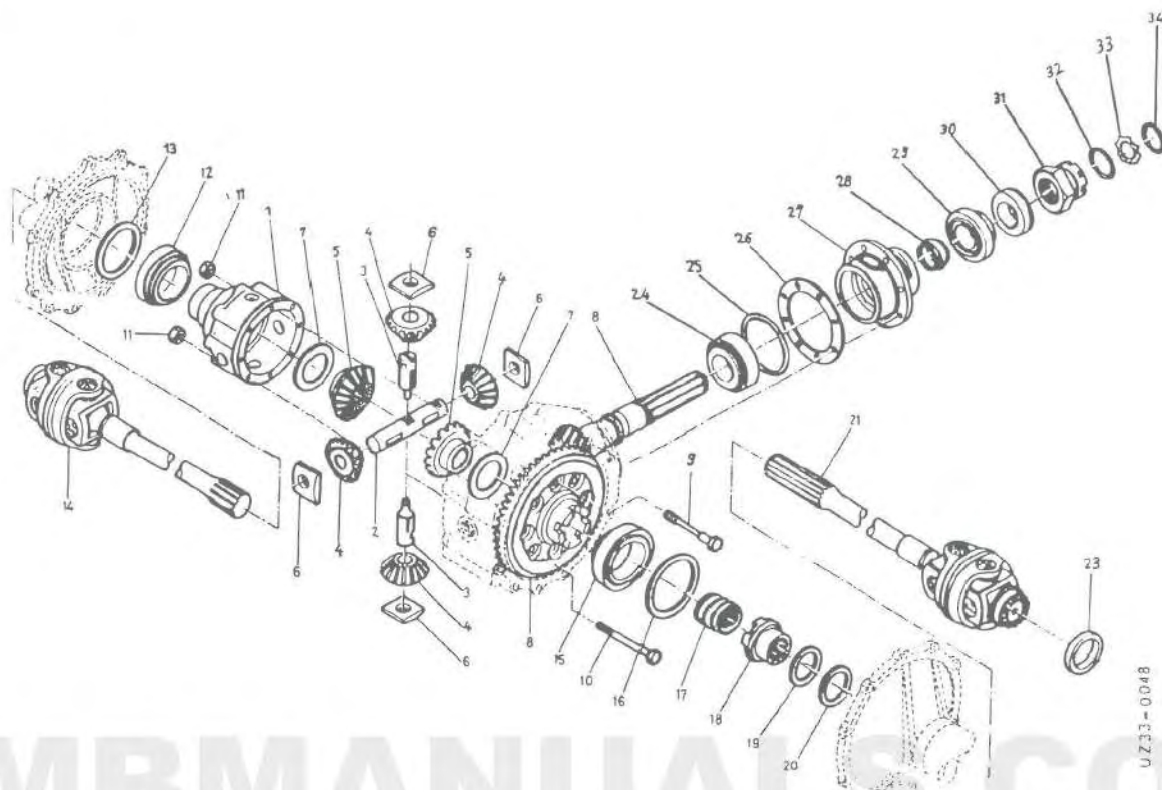
Axle tubes and steering knuckles

1	Drive housing	24	Steering knuckle
2	Screw plug	25	Bolt
5	Axle tube	26	Swivel joint
6	Needle bearing	27	Sealing ring
7	Snap ring	28	Bushing
8	Sealing ring	29	Plate
9	Axle tube	30	Washer
10	Sealing ring	31	Bolt
11	Needle bearing	32	Grease nipple
12	Snap ring	33	Sliding block
13	Piston	34	Thrust shell
14	Sealing ring	35	Sealing ring
15	Sealing ring	36	Wear ring
16	Bolt	37	Sealing ring
18	Sealing ring	39	Bolt
19	Socket	40	Bolt
20	Washer	41	Bolt
21	Bolt	42	Bolt
22	Nut	43	Grease nipple
23	Bolt		

Text and illustration do not agree with microfiche.

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Exploded View

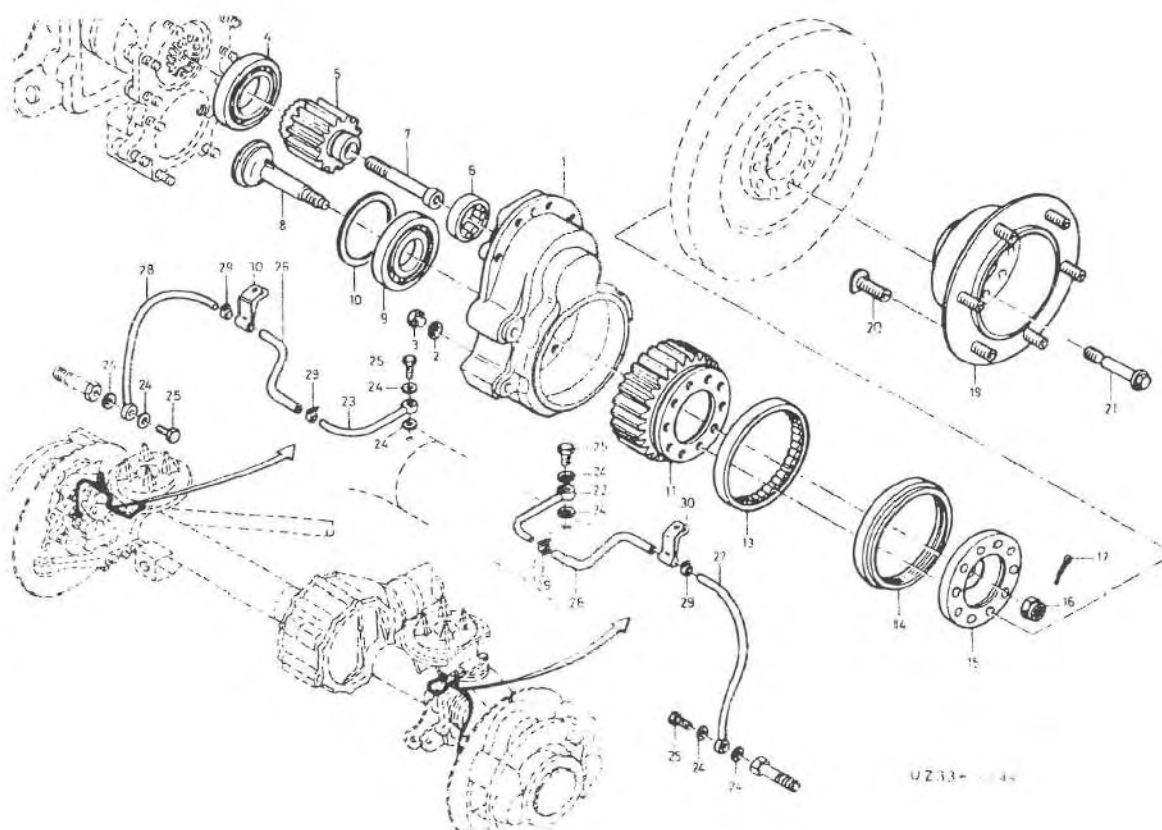


Differential

- | | |
|-----------------------------|-------------------------|
| 1 Differential housing | 18 Shift dog |
| 2 Differential pinion shaft | 19 Thrust washer |
| 3 Differential pinion shaft | 20 Thrust washer |
| 4 Differential pinion | 21 Drive shaft |
| 5 Side gear | 23 Wear ring |
| 6 Thrust washer | 24 Taper roller bearing |
| 7 Thrust washer | 25 Sealing ring |
| 8 Gear set | 26 Shim |
| 9 Bolt | 27 Pinion flange |
| 10 Bolt | 28 Spacing sleeve |
| 11 Nut | 29 Taper roller bearing |
| 12 Taper roller bearing | 30 Wear ring |
| 13 Shim | 31 Nut |
| 14 Drive shaft | 32 Sealing ring |
| 15 Taper roller bearing | 33 Retainer |
| 16 Shim | 34 Lock washer |
| 17 Spring | |

Text and illustration do not agree with microfiche.

Exploded View



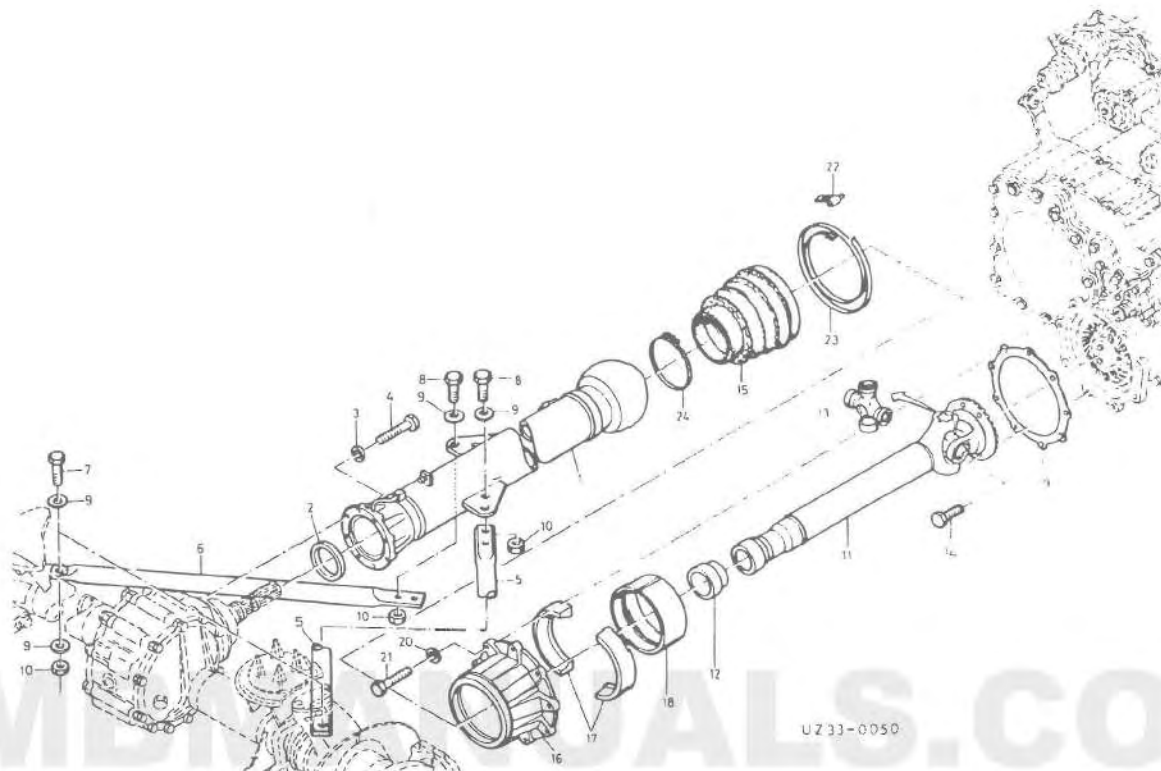
Wheel hub drive and wheel hub

1	Housing	16	Nut
2	Sealing ring	17	Split pin
3	Bolt	19	Wheel hub
4	Ball bearing	20	Wheel stud
5	Drive gear	21	Bolt
6	Cylindrical roller bearing	22	Vent pipe
7	Bolt	23	Vent pipe
8	Bolt	24	Sealing ring
9	Ball bearing	25	Union screw
10	Shim	26	Hose
11	Driven gear	27	Line
13	Cylindrical roller bearing	28	Line
14	Shaft seal	29	Clamp
15	Wear ring	30	Clamp

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Exploded View



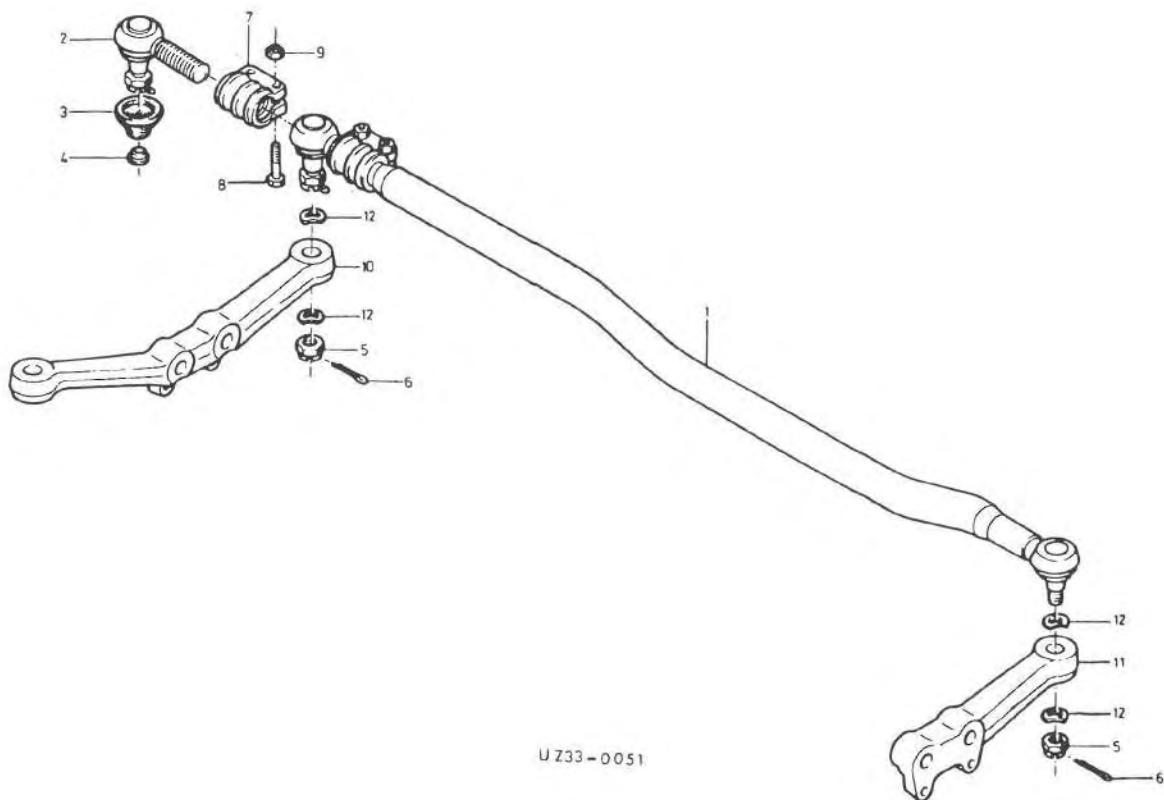
Torque tube

- | | |
|--------------------|-----------------------|
| 1 Torque tube | 13 Spider |
| 2 Sealing ring | 14 Bolt |
| 3 Washer | 15 Rubber gaiter |
| 4 Bolt | 16 Torque ball casing |
| 5 Axle strut | 17 Torque ball shell |
| 6 Axle strut | 18 Torque ball shell |
| 7 Bolt | 19 Shim |
| 8 Bolt | 20 Washer |
| 9 Washer | 21 Bolt |
| 10 Nut | 22 Expanding clip |
| 11 Propeller shaft | 23 Strap |
| 12 Bushing | 24 Clamp |

Text and illustration do not agree with microfiche.

737.111

Exploded View



Track rod and steering knuckle arm

- 1 Track rod
- 2 Ball joint
- 3 Collar
- 4 Clamping ring
- 5 Nut
- 6 Split pin
- 7 Clamp
- 8 Bolt
- 9 Nut
- 10 Steering knuckle arm
- 11 Steering arm
- 12 Washer

Text and illustration do not agree with microfiche.

Work Sheet for Adjusting Axle Drive

Axle Models 737.0 – 747.0/737.1 – 747.1

UNIMOG – MB trac



Customer _____

Dealer _____

Date _____

Order No. _____

Registration No. _____

Initial registration _____

Chassis No. _____

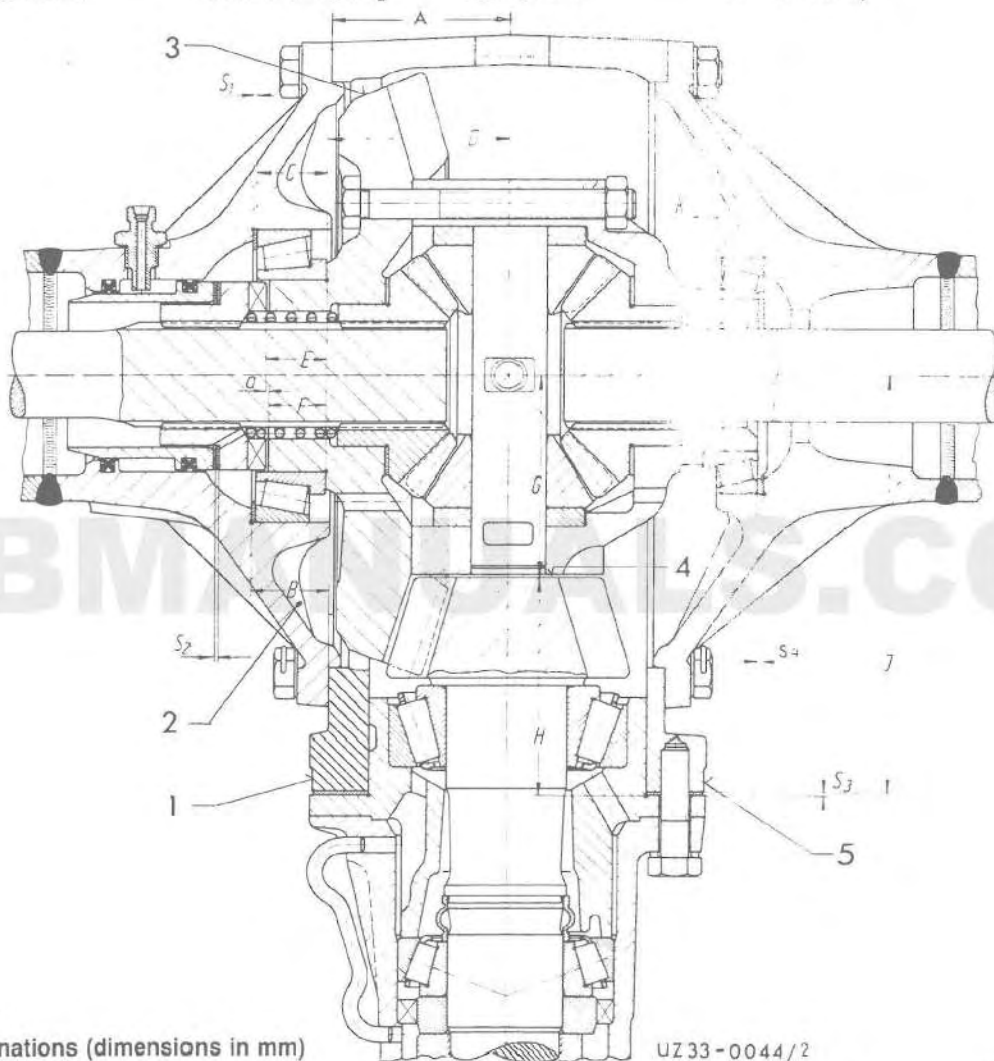
Axle No. _____

Service hours _____

Odometer reading _____

Last service _____

Carried out by _____



Explanations (dimensions in mm)

UZ 33-0044/2

A	Etched in center axle housing (1) ¹⁾	e.g.	90
B	Stamped in axle tube at lock side (2)	e.g.	33,15
C	Overall height of large taper roller bearing, measured on a surface plate	e.g.	29,00
D	Etched in ring gear (3)	e.g.	76,90
E	Measurement from collar of inner bearing race to upper edge of shift dog (washer (s) S ₁ , taper roller bearing, piston, thrust washer and shift dog installed in axle tube); measured with depth gauge		
F	Measurement from upper claw edge ring gear to contact surface of bearing at ring gear	e.g.	34,80
G	Measurement from end face of drive pinion to center differential etched in end face of drive pinion (4)	e.g.	33,30
H	Measurement from end face of drive pinion flange;		
J	measured with depth gauge	e.g.	86,15
K	Measurement from center of differential to flange is stamped in top side of axle	e.g.	92,05
L	at center of axle housing (5) ²⁾		
	Measurement from inner bearing race to contact surface of bearing at differential housing	e.g.	77
	Thickness of Nilos ring, measured with micrometer		

- Important!** All readings differ from axle to axle.
 For housing width 130 mm, insert the constant 64 before the measurement
 (e.g. 90); and the constant 74 for housing width 140 mm.
 2) Insert the constant 176 before the measurement (e.g. 77); and constant 177 for dimension 00.

33.6 General

737.111

Break-away torque for adjustment of drive pinion with oiled taper roller bearings (SAE 90).

	003 981 92 05	Specified = 3,0 to 3,5
Taper roller bearing	+ 004 981 28 05	Nm ¹⁾
	001 981 46 05	
	+ 001 981 44 05	Actual =

Important: Rinse out taper roller bearing and insert with gear oil SAE 90.

Note readings and calculate shim thickness.

- Shim thickness S_1 for large taper roller bearing at lock side determines backlash and contact pattern together with shim thickness S_3 .

$$S_1 = (A + B) - (C + D)$$

A =	C =	(A + B) =
+ B =	+ D =	- (C + D) =
(A + B) =	(C + D) =	S ₁ =

- Shim thickness S_2 between piston and differential lock determines claw play (a) of differential lock at off position.

$$S_2 = E - (F + a)$$

E =	F =	(a = claw play 0,5±0,2 mm)
- (F + a) =	+ a =	
S ₂ =	F + a =	

- Shim thickness S_3 at flange of drive pinion bearing determines amount by which drive pinion engages ring gear.

$$S_3 = (G + H) - J$$

G =	(G + H) =
+ H =	- J =
(G + H) =	S ₃ =

- Shim thickness S_4 at small taper roller bearing for front axle and rear axle.

$$S_4 = K$$

Explanations to shim thickness S_4 .

The coil spring of the differential lock between the shift dog and ring gear is **not** to be fitted for measurement.

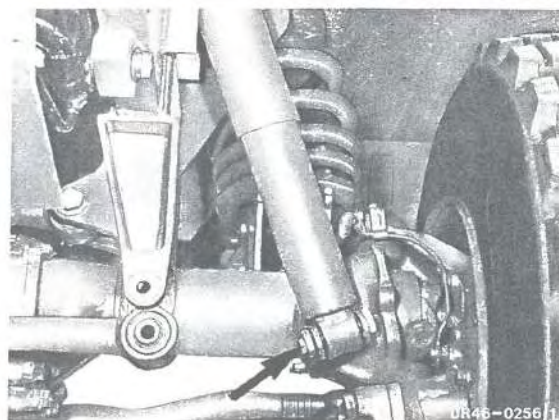
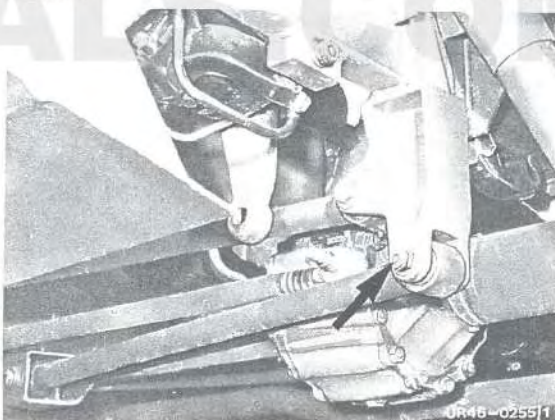
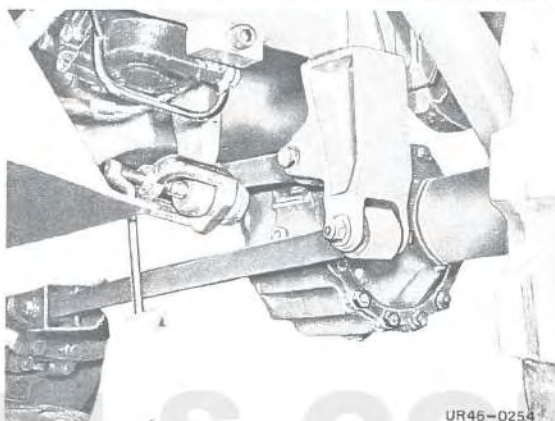
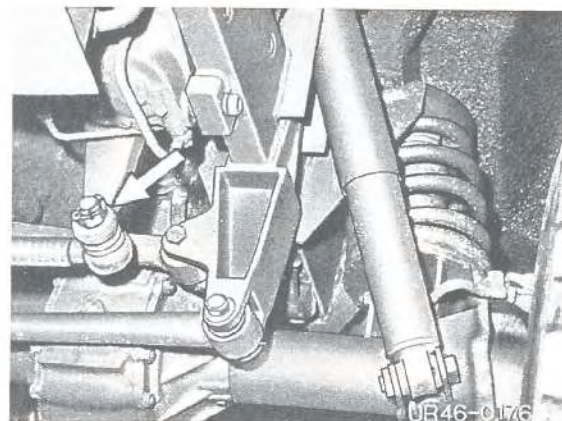
- 4.1 Force outer bearing race into axle tube without shims.
- 4.2 Force taper roller bearing with inner race onto bearing seat of differential housing by half the bearing thickness.
- 4.3 Introduce differential into center axle housing at lock side.
- 4.4 Position axle tube on center axle housing and torque **four opposite bolts** to 135 Nm²⁾ while steadily turning ring gear.
- 4.5 After tightening bolts, continue to rotate ring gear while knocking (plastic-headed hammer) flange part of axle tube to eliminate stresses from bearing seat.
- 4.6 Remove axle tube and withdraw outer bearing race again.
- 4.7 Measure gap K with feeler gauge, insert shims S_4 of calculated thickness beneath outer race of taper roller bearing, and force race into position.
- 4.8 Take differential out.
- 4.9 Fully press inner race of taper roller bearing onto bearing seat of differential housing.

- 1) Taper roller bearing 001 981 44 05 Specified = 2,0 to 2,2 Nm
+ 001 981 37 05

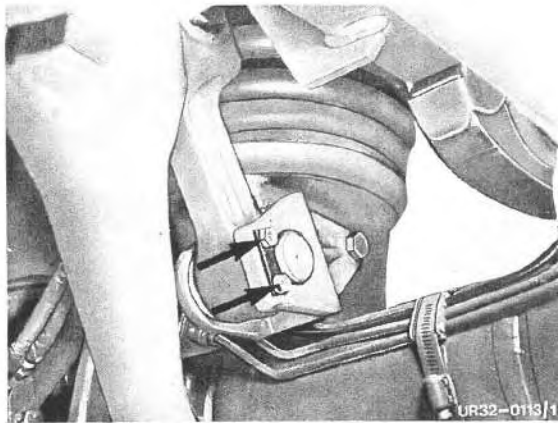
- 2) For axle model 737/747.111 torque to 200 Nm.

Removal

- 1 Remove spare wheel.
- 2 Release and unscrew castle nut at drag link.
- 3 Using special tool No. 8, pull ball joint off pitman arm.
- 4 Unscrew lower transverse link.
- 5 Unscrew shock absorber at axle.

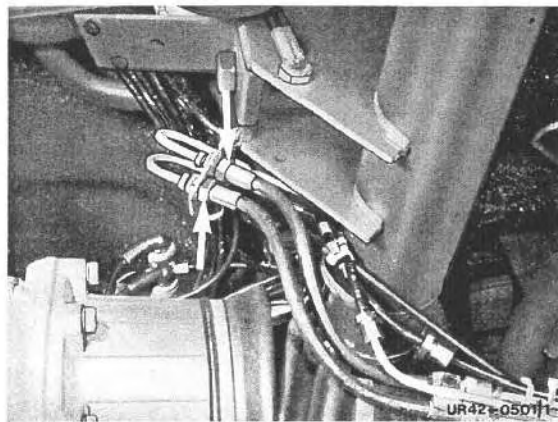


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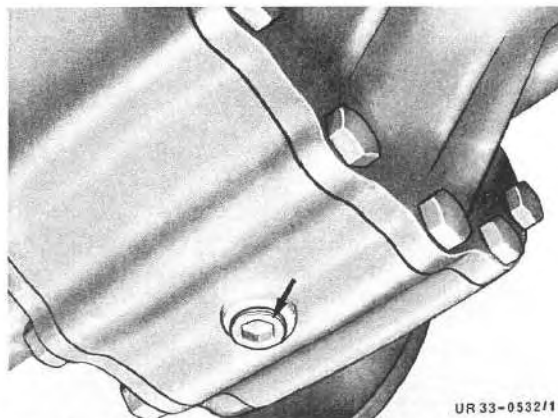


6 Unscrew stabilizer retainers at left and right, remove pins.

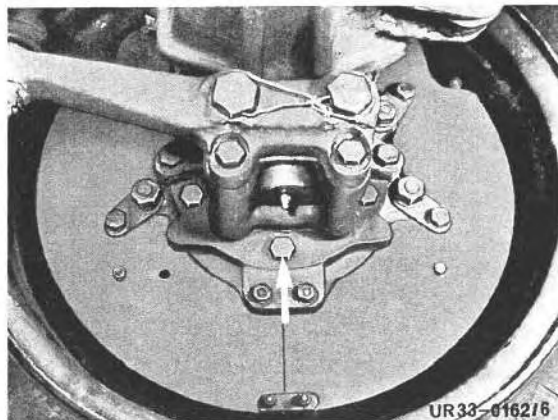
Note: Detach wheel if pins are stuck.



7 Then unscrew air supply and exhausting lines, differential lock line, brake shoe wear indicator cable and front brake circuit lines. Close ends of all brake lines.

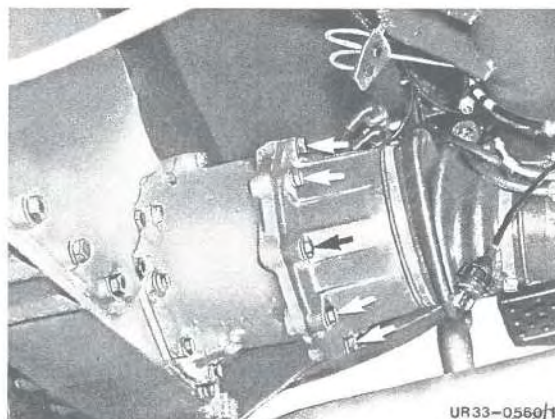


8 Drain oil off center axle housing.



9 Drain oil off wheel hub drive.

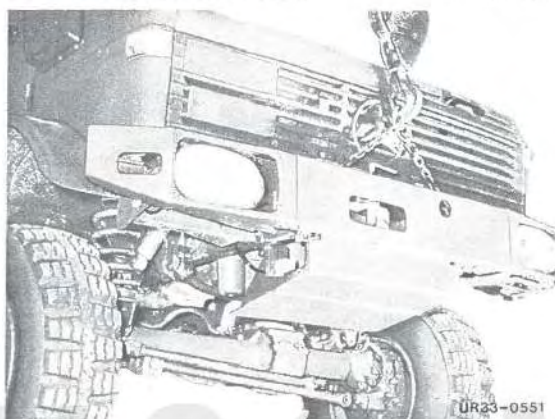
- 10 Unscrew torque ball housing.



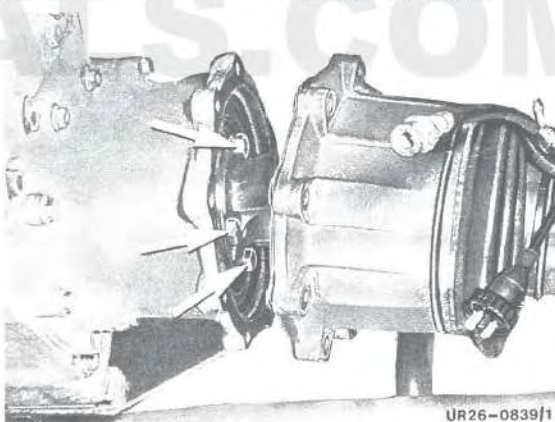
- 11 Raise vehicle at front, push axle forward and support torque tube.

Note: To raise vehicle, insert chain lifting tackle at both front ends of frame.

- 12 Remove front half shells of torque ball.

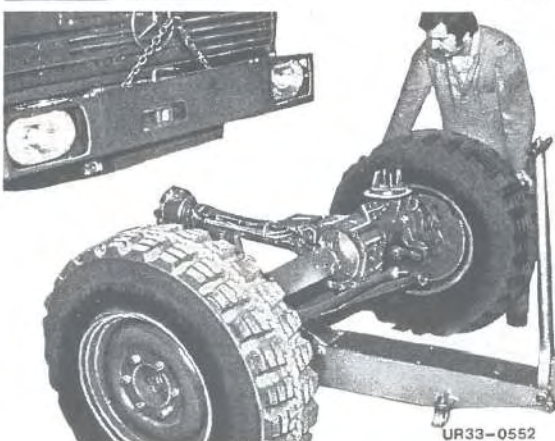


- 13 Unscrew propeller shaft from transmission.

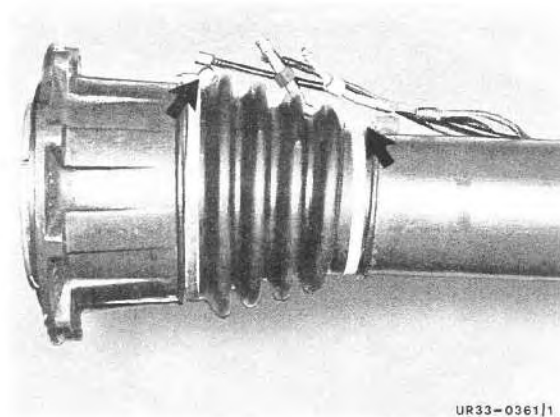


- 14 Lift vehicle out of springs and move axle forward.

- 15 Lower vehicle onto supporting pedestals.

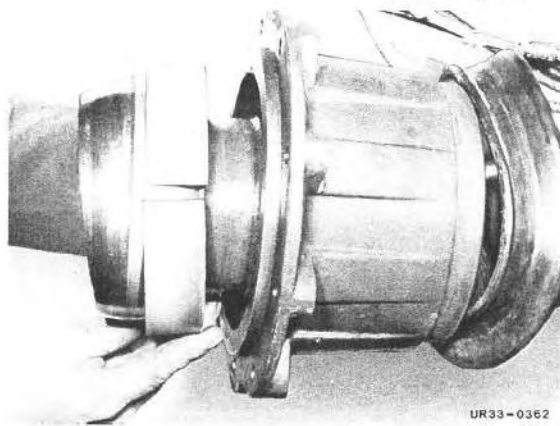


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UR33-0361/1

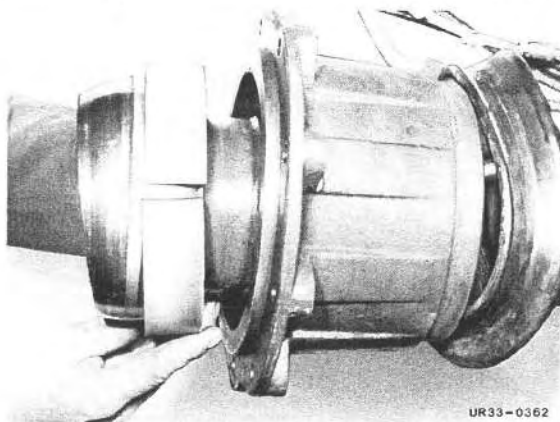
16 Remove hose bands and pull back rubber sleeve.



UR33-0362

17 Pull back torque ball housing and remove rear torque ball half shells.

18 Clean torque ball half shells and torque ball housing, check and replace if necessary.



UR33-0362

Preparing for installation

1 Carefully deburr torque ball housing and half shells of torque ball.

2 Coat rear half shells of torque ball with paste No. 3 and install.

3 Coat front half shells of torque ball with paste No. 3 and install.

Note:

From Model	Chassis end No.	Front axle end No.
435.115	081 106	004 567
.117		

The shims, the torque ball housing and bearing shells of the torque balls were modified as follows:



UR 33-0195